

AWS Graviton vs. AMD EPYC vs. Intel Xeon: A Comparative Architecture Analysis of Modern Server Processors

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Abstract

This document reports the progress made on the term paper titled *AWS Graviton vs. AMD EPYC vs. Intel Xeon: A Comparative Architecture Analysis of Modern Server Processors* as of 24 August 2026. Per the instructions for Draft 1, this document is a **progress report** — the “Background Preparations” expected at this stage — and not a preliminary version of the final ~150-page term paper.

The chosen topic undertakes a comparative architectural study of three families of modern data-centre server processors: Amazon Web Services’ custom Arm-based Graviton processors, AMD’s x86-64 EPYC processors (Zen microarchitecture family), and Intel’s x86-64 Xeon Scalable processors. The comparison spans core microarchitecture, cache and memory-system design, multicore/multiprocessor organisation, interconnect topology, and power/performance trade-offs — all of which map onto Chapter 12 (*Multiprocessor Systems*), Chapter 11 (*The Memory System*), and Appendix A (*Case Studies of Real Processors*) of the prescribed textbook.

This draft records the selected topic, the textbook mapping, the identified work packages, literature-survey progress so far, preliminary technical understanding, a Gantt chart of progress, and the plan for subsequent cumulative drafts. The detailed quantitative benchmarking, the full microarchitectural comparison, and the complete literature synthesis will be developed in later submissions.

Keywords: server processors, AWS Graviton, AMD EPYC, Intel Xeon, multicore architecture, cache coherence, memory system, Arm Neoverse, x86-64, data-centre computing

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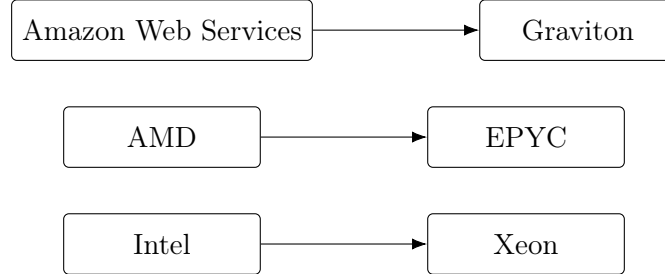
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1 Work Done as of 24 August 2026

1.1 What is My Chosen Topic

The chosen topic is a comparative architecture analysis of three server-class processor families used in modern data centres and cloud platforms:



The three families differ at the instruction-set level (Graviton implements Instruction Set Architecture (ISA) ARM; EPYC and Xeon implement x86-64) and further in microarchitecture, cache hierarchy, multicore organisation, interconnect, and I/O design. The paper studies how these architectural choices affect performance, scalability, and energy efficiency.

To keep the comparison tractable, the team has decided to focus on one representative modern generation per family rather than every generation ever released: **AWS Graviton4**, **AMD EPYC 9005 (Turin)**, and **Intel Xeon 6 (Granite Rapids)**. These three generations were selected because prior same-cloud, same-instance-family benchmarking of earlier generations from all three vendors has already shown this kind of controlled comparison is possible [6]. [6] itself only covers Graviton/Graviton2/Graviton3, EPYC Milan, and Xeon Ice Lake — the team still needs to locate (or produce) an equivalent same-platform benchmark specifically for Graviton4, EPYC 9005 (Turin), and Xeon 6 (Granite Rapids).

1.2 Why the Topic was Selected

The server processor landscape is undergoing a fundamental shift. For decades, x86 processors from Intel and AMD dominated data centres. The emergence of AWS Graviton — custom ARM-based silicon optimised for cloud workloads — challenges the hegemony of x86 and raises deep architectural questions about ISA efficiency, power-performance trade-offs, and vertical integration. This topic directly applies almost every major concept taught in ELL305 (ISA design, pipelining, memory hierarchy, multiprocessors) to real, commercially deployed hardware, making it both academically rigorous and practically relevant.

1.3 Mapping to the Prescribed Textbook

Textbook: *Basic Computer Architecture*, Smruti R. Sarangi, Version 3.09, 2025.

URL: <https://srsarangi.github.io/archbook/archbook.pdf>

The paper spans the following chapters, sections, and sub-sections, as summarised in Table 1:

Faculty consultation. As this topic is drawn from Chapters 1–12, the team will consult **Prof. Sumantra DuttaRoy** for questions on the topic and its textbook mapping, per

Chapter / Section	Relevance to the Term Paper
Ch. 1, §1.5 — Instruction Set Design	RISC vs. CISC distinction; motivates the ARM64 (Graviton) vs. x86-64 (EPYC, Xeon) comparison.
Ch. 4 — ARM Assembly Language	ISA-level foundation for Graviton’s ARM64 execution model, register file, and addressing modes.
Ch. 5 — x86 Assembly Language	ISA-level foundation for EPYC and Xeon; CISC encoding, micro-op translation.
Ch. 9 — Processor Design	Fetch/Decode/Execute/MA/RW pipeline stages; control unit; hard-wired vs. microprogrammed design — vocabulary for comparing all three chips.
Ch. 10 — Principles of Pipelining	§10.4 hazards, §10.9 performance metrics (CPI, IPC), §10.10 power/temperature (ED^2), §10.11 branch prediction, multiple-issue, and out-of-order execution.
Ch. 11 — The Memory System	§11.2 caches, §11.3 miss penalties, §11.4 virtual memory and MMU — used to compare L1/L2/L3 hierarchies, NUMA topology, and virtualisation support.
Ch. 12 — Multiprocessor Systems	§12.1 Moore’s Law, §12.4 MIMD/coherence/consistency, §12.4.4 UMA vs. NUMA, §12.6 interconnection networks (Infinity Fabric, Intel UPI, Graviton fabric).
Appendix A — Case Studies of Real Processors	AMD Bulldozer (§A.2.2) and Intel Sandy Bridge (§A.3.2) serve as the methodological template for our processor case-study chapters.

Table 1: Textbook coverage map for the term paper.

the course rule that Chapters 1–12 route to Prof. DuttaRoy and Chapter 13 routes to Prof. Sarangi.

1.4 Progress So Far (Gantt Chart)

The chart below shows the work completed or currently in progress up to **24 August 2026**. It is intended as a record of progress made so far rather than a detailed plan for the future. The detailed schedule for subsequent drafts and the final submission will be developed in later drafts.

Work-package breakdown

- **WP1 – Textbook Study.**

- Task 1: Read the basic chapters.
 - * Read Ch. 1 (Introduction), Ch. 4 (ARM ISA), and Ch. 5 (x86 ISA) — **70%** done by 24 Aug 2026.
 - * Read Ch. 9 (Processor Design) and Ch. 10 (Pipelining) — **50%** done.
 - * Read Ch. 11 (Memory System), Ch. 12 (Multiprocessors), and Appendix A — **30%** done.
- Task 2: Read papers related to the topic and store the references and PDFs in Zotero.

- **WP2 – Literature.**

- Task 3: 9 references identified and collected so far (full list given in the following subsection).

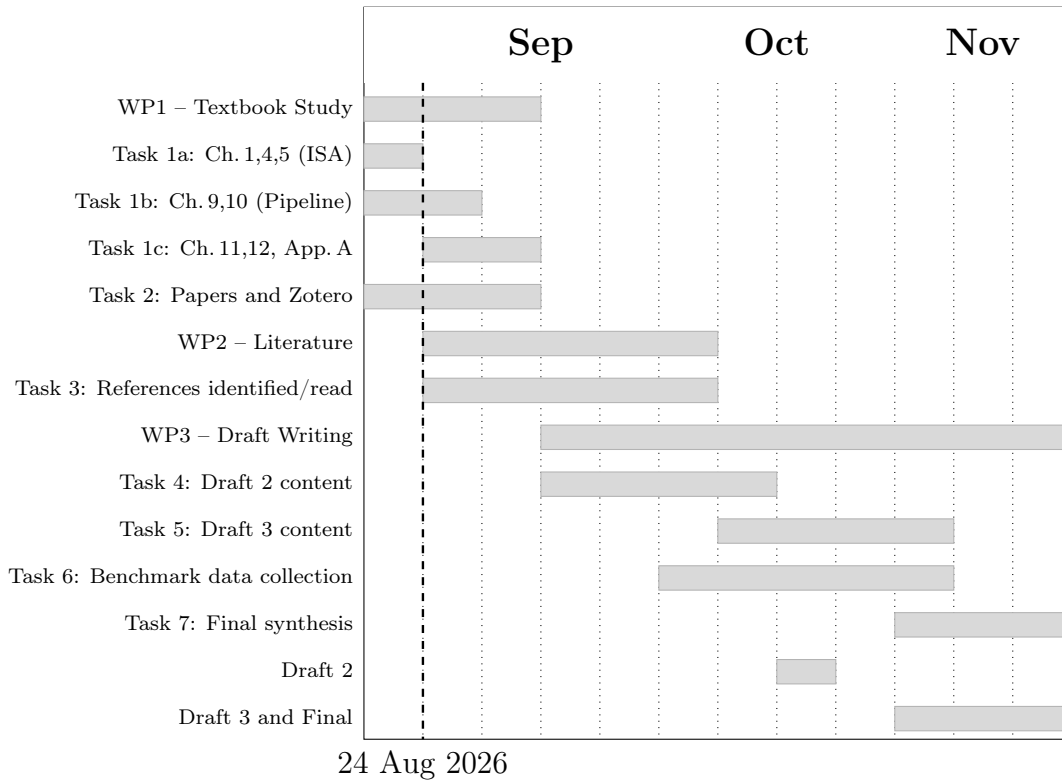


Figure 1: Gantt chart showing the textbook study, literature collection, paper reading, drafting, benchmarking, and final synthesis activities. The dashed vertical line marks the current reporting date, 24 August 2026.

1.5 Preliminary Technical Understanding

ISA Comparison: ARM64 vs. x86-64

AWS Graviton implements the **ARMv8.x/ARMv9** ISA — a *Reduced Instruction Set Computer (RISC)* architecture with fixed 32-bit instruction width, 31 general-purpose

registers, and a load/store-only memory model. EPYC and Xeon implement **x86-64** — a *Complex Instruction Set Computer (CISC)* ISA with variable-length encoding (1–15 bytes), 16 architectural registers, and memory operands in arithmetic instructions. Xeon and EPYC both translate x86 macro-ops into internal RISC-like micro-ops before the out-of-order engine, adding a decode step absent in Graviton.

Microarchitecture of the Three Chosen Generations

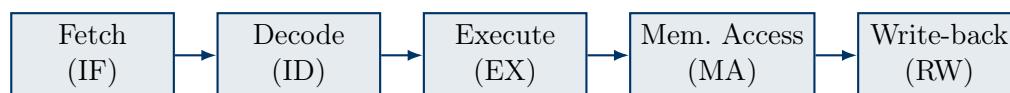
AWS Graviton4 (2024) Neoverse V2 core; 96 cores per socket; foundry/process node not publicly disclosed by AWS; SVE2 at a 128-bit vector length (four 128-bit pipelines, *not* 512-bit as sometimes assumed); 6-wide decode, 8-wide dispatch [1].

AMD EPYC 9005 “Turin” (2024) Core Complex Die (AMD chiplet containing CPU cores) (CCD) Zen 5 (standard) or Zen 5c (dense) cores; up to 128 cores (Zen 5: 16 CCDs $\times$ 8 cores, TSMC 4 nm) or up to 192 cores (Zen 5c: 12 CCDs $\times$ 16 cores, TSMC 3 nm), plus a separate I/O die; novel dual 4-wide decode clusters, up to 8-wide combined with Simultaneous Multithreading (SMT) [2, 3, 7].

Intel Xeon 6900P “Granite Rapids” (2024) Redwood Cove P-core; up to 128 cores per socket; Intel 3 process; Advanced Matrix Extensions (AMX) support; front end inherited from Golden Cove’s 6-wide decode [4, 5].

Pipeline Design and Out-of-Order Execution

All three processors are deeply superscalar, out-of-order execution designs, each built around Tomasulo-style reservation stations, a Reorder Buffer (ROB) for precise exceptions, and aggressive branch prediction — concepts grounded in §10.11 of the textbook.



Reservation stations + reorder buffer (ROB)
sit around EX for out-of-order issue and precise
exceptions (Graviton4, Zen 5, Redwood Cove).

Figure 2: Generic five-stage in-order front-end / out-of-order back-end pipeline (Ch. 9–10) used as the reference model for comparing the three processors.

Cache Hierarchy

Multi-core and Multi-chip Interconnects

- **EPYC:** AMD Infinity Fabric — die-to-die coherence across up to 16 CCDs; NUMA topology.
- **Xeon:** Intel Ultra Path Interconnect (UPI) for multi-socket; mesh-based on-die interconnect for tile coherence.
- **Graviton:** AWS proprietary on-package fabric; single-socket design targeting TCO over raw throughput.

Level	Graviton4	EPYC 9005 (Turin)	Xeon 6900P (Granite Rapids)
L1-I	64 KB	32 KB	32 KB
L1-D	64 KB	48 KB	48 KB
L2	2 MB	1 MB	2 MB
L3 (shared, flagship SKU)	36 MB (96 cores)	up to 512 MB (128 cores)	up to 504 MB (128 cores)

Table 2: Cache hierarchy of the flagship SKU per family (per-core unless noted). Sources: Graviton4 [1]; EPYC 9005 [2, 3]; Xeon 6900P [4, 5].

Virtualisation

All three support hardware virtualisation: ARM VHE/VHE+NV2, AMD-V with SEV-SNP, and Intel VT-x with TDX. AWS’s Nitro System additionally offloads hypervisor functions to dedicated ASICs, reducing virtualisation overhead in a way that is unique to Graviton-based instances.

1.6 References Read So Far

- Sarangi [8] — prescribed textbook
- Velten et al. [9] — experimentally compares the memory hierarchy of AMD EPYC Rome and Intel Cascade Lake SP.
- AWS [10] — official AWS Graviton technical guide (architecture, tooling, migration notes).
- AMD [2] — AMD EPYC 9005 (Turin) architecture overview, doc. 58462, Rev. 1.0.
- Intel [4] — Intel Xeon 6 architecture support article on performance and efficiency cores.
- Larabel [6] — Phoronix same-cloud benchmark of Graviton/Graviton2/Graviton3 vs. EPYC Milan vs. Xeon Ice Lake (earlier generations; see the flag in Section 1.1).
- Lawenda et al. [11] — effect of AMD EPYC L3 cache size on CFD application performance.
- Hofmann et al. [12] — Execution-Cache-Memory performance model for predicting memory-bound kernel performance.
- AWS [13] — AWS whitepaper on Graviton performance-testing methodology for ISVs.

A Plagiarism Report

The text of this submission was checked using the Grammarly Plagiarism Checker on 24 August 2026. The report found that **2%** of the submitted text matches external sources available on the web or in academic databases.

The identified matches were reviewed to distinguish legitimate source-based material from unattributed copying. The matching material primarily corresponds to technical information and source-related content used in the literature survey. Appropriate citations have been provided for material derived from external sources.

Plagiarism Check Parameter	Result
Software Used	Grammarly Plagiarism Checker
Date of Check	24 August 2026
External-Source Similarity	2%
AI-Pattern Indicator	8%

Table 3: Results of the plagiarism and AI-pattern check.

B Reading Indices

Table 4 reports standard readability indices (see https://www.analyzemywriting.com/readability_indices.html) computed on the prose of Sections 1.1–1.6 of this draft.

Index	Score
Gunning Fog Index	17.86
SMOG Index	15.57
Coleman–Liau Index	18.5
Automated Readability Index	16.1

Table 4: Readability indices computed on this draft’s prose.

The uniformly high grade-level scores are expected for a computer-architecture term paper: roughly a quarter of the words in this draft are technical multi-syllable terms (e.g. “microarchitecture”, “instruction-set”, “virtualisation”), which every one of the indices above penalises regardless of how clearly the sentences are constructed.

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Glossary

cache coherence The property that all cores in a multicore system observe a single, consistent view of any given memory location despite each core holding its own cached copy

out-of-order execution Executing instructions in an order determined by operand availability rather than strictly the original program order, while preserving the illusion of in-order execution

pipeline An implementation technique in which multiple instructions are overlapped in execution, each moving through a fixed sequence of stages (fetch, decode, execute, memory access, write-back)

superscalar A processor design capable of issuing and executing more than one instruction per clock cycle using multiple parallel execution units

Acronyms

AMX Advanced Matrix Extensions

CCD Core Complex Die (AMD chiplet containing CPU cores)

CISC Complex Instruction Set Computer

ISA Instruction Set Architecture

NUMA Non-Uniform Memory Access

RISC Reduced Instruction Set Computer

ROB Reorder Buffer

SMT Simultaneous Multithreading

UPI Ultra Path Interconnect

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